PRACTICE QUESTIONS CONTINUOUS RANDOM VARIABLES

EXERCISE – 3.2

PROBABILITY DENSITY FUNCTIONS

1. The continuous random variable X has probability density function $f(x)$ where

$$f(x)=\begin{cases} kx^2 & 0\le x\le 2 \\ 0 & otherwise \end{cases}$$

Find the following
(a) value of constant k
(b) $P(X\ge 1)$
(c) $P(1/2<X<3/2)$
$(Ans: 3/8, 7/8, 13/32)$

2. The continuous random variable X has probability density function $f(x)$ where

$$f(x)=\begin{cases} k(4-x) & 1\le x\le 3 \\ 0 & otherwise \end{cases}$$

Find the following
(a) value of constant k
(b) $P(1.2\le X\le 2.4)$
$(Ans: 0.25, 0.66)$

3. A continuous random variable X has p.d.f $f(x)$ where

$$f(x)=\begin{cases} A(2-x)(2+x) & 0\le x\le 2 \\ 0 & otherwise \end{cases}$$

Find the following
(a) value of constant A
(b) $P(X\le 1)$ (c) $P(X\ge 2)$ (d) $P(1<X<2)$ (e) $P(X=1/2)$
$(Ans: 3/16, 11/16, 0, 5/16, 0)$

4. A continuous random variable X has p.d.f $f(x)$ where

$$f(x)=\begin{cases} x & 0<x<1 \\ 2-x & 1<x<2 \\ 0 & otherwise \end{cases}$$

(a) Show that $P(0\le X\le 2)=1$
(b) Find $P(X<1.2)$
(c) Find $P(X\ge 0.5)$
$(Ans: 0.68, 0.875)$

HAZBER SAMSON

5. A continuous random variable X has p.d.f $f(x)$ where

$$f(x)=\begin{cases} k & 0\le x<2 \\ k(2x-3) & 2\le x\le 3 \\ 0 & otherwise \end{cases}$$

Find the following
(a) value of constant *k*
(b) $P(X\le 1)$
(c) $P(X>2.5)$
(d) $P(1\le X\le 2.3)$
(e) $P(X=2)$
$(Ans: 0.25, 0.25, 0.3125, 0.3475, 0)$

DISTRIBUTION FUNCTIONS FOR CONTINUOUS RANDOM VARIABLE

6. The continuous random variable X has probability density function $f(x)$ where

$$f(x)=\begin{cases} \frac{3}{8}x^2 & 0\le x\le 2 \\ 0 & otherwise \end{cases}$$

Find the cumulative distribution function

$$(Ans: F(x)=\begin{cases} 0 & x<0 \\ \frac{x^3}{8} & 0\le x\le 2) \\ 1 & x\ge 2 \end{cases}$$

7. The continuous random variable X has probability density function $f(x)$ where

$$f(x)=\begin{cases} \frac{1}{4} & 0\le x\le 2 \\ \frac{1}{4}(2x-3) & 2\le x\le 3 \\ 0 & otherwise \end{cases}$$

Find the cumulative distribution function

$$(Ans: F(x)=\begin{cases} 0 & x<0 \\ \frac{x}{4} & 0\le x\le 2 \\ \frac{1}{4}(x^2-3x+4) & 2\le x\le 3 \\ 1 & x\ge 3 \end{cases}$$

HAZBER SAMSON

8. The continuous random variable X has probability density function $f(x)$ where

$$f(x)=\begin{cases} c/\sqrt{x} & 0\le x\le 4 \\ 0 & otherwise \end{cases}$$

Find the following
(a) the value of c (b) The distribution function of X
(c) $P(X>1)$ (d) $P(2\le X\le 3)$

$$(Ans: c=1/4,\ F(x)=\begin{cases} 0 & x<0 \\ \dfrac{\sqrt{x}}{2} & 0\le x\le 4, \\ 1 & x\ge 4 \end{cases})(0.5, 0.159)$$

9. The continuous random variable X has the distribution function

$$F(x)=\begin{cases} 0 & x<0 \\ \dfrac{x^3}{27} & 0\le x<3 \\ 1 & x\ge 3 \end{cases}$$

Find the following
(a) The p.d.f of X (b) $P(X\le 2)$ (c) $P(1\le X\le 2)$

$$(Ans: p.d.f\ f(x)=\begin{cases} x^2/9 & 0\le x\le 3 \\ 0 & otherwise \end{cases}$$

$(Ans: 8/27, 7/27)$

10. The continuous random variable X has the distribution function

$$F(x)=\begin{cases} 0 & x<0 \\ \dfrac{2x^2}{5} & 0\le x<1 \\ -\dfrac{3}{5}+\dfrac{2}{5}\left(3x-\dfrac{x^2}{2}\right) & 1\le x<2 \\ 1 & x\ge 2 \end{cases}$$

Find the following
(a) The p.d.f of X
(b) $P(|X|>1.25)$

$$(Ans: p.d.f\quad f(x)=\begin{cases} \dfrac{4x}{5} & 0\le x<1 \\ 2(3-x)/5 & 1\le x<2 \\ 0 & otherwise \end{cases}$$

$(Ans: 0.4125)$

HAZBER SAMSON

EXPECTATION AND VARIANCE OF CONTINUOUS RANDOM VARIABLE

11. Find $E(X)$ for each of the following continuous random variables.

(a) $f(x)=\begin{cases}\frac{3}{4}(x^2+1) & 0\le x\le 1\\ 0 & otherwise\end{cases}$

(b) $f(x)=\begin{cases}\frac{3}{4}x(2-x) & 0\le x\le 2\\ 0 & otherwise\end{cases}$

(c) $f(x)=\begin{cases}\frac{3}{8} & \frac{2}{3}\le x<2\\ \frac{3}{32}x(4-x) & 2\le x\le 4\\ 0 & otherwise\end{cases}$

$(Ans: 9/16, 1, 49/24)$

12. If X is a continuous random variable with p.d.f $f(x)$. Find $E(X)$

(a) $f(x)=\begin{cases}k\,x^2 & 0\le x\le 4\\ 0 & otherwise\end{cases}$

(b) $f(x)=\begin{cases}k\,x^3 & 0\le x\le 2\\ 0 & otherwise\end{cases}$

$(Ans: 3, 1.6)$

13. The continuous random variable X has p.d.f $f(x)$ where

$$f(x)=\begin{cases}k\,x & 0\le x<1\\ k & 1\le x<3\\ k(4-x) & 3\le x\le 4\\ 0 & otherwise\end{cases}$$

(a) Find k

(b) Find $E(X)$

$(Ans: 1/3, 2)$

14. A random variable X has probability density function $f(x)$ given by

$$f(x)=\begin{cases}c\,x(5-x) & 0\le x\le 5\\ 0 & otherwise\end{cases}$$

Show that $c=6/125$ and find the mean of X. $(Ans: 2.5)$

HAZBER SAMSON

15. A random variable X has probability density function $f(x)$ given by

$$f(x)=\begin{cases}\frac{c}{3}x & 0\le x<3\\ c & 3\le x\le 4\\ 0 & otherwise\end{cases}$$

Than find
(a) the value of c
(b) the mean of X
(c) the value of a for there to be a probability of 0.85 that a randomly observed value of X will exceed a.
$(Ans: 0.4, 2.6, 1.5)$

16. Given the p.d.f of the function is denoted by $f(x)$. Than find the following
(a) $E(X)$ (b) $E(X^2)$ (c) $Var(X)$ (d) standard deviation of X for

(a) $f(x)=\begin{cases}\frac{3}{8}x^2 & 0\le x\le 2\\ 0 & otherwise\end{cases}$

$(Ans: 1.5, 2.4, 0.15, 0.387)$

(b) $f(x)=\begin{cases}\frac{1}{4}(4-x) & 1\le x\le 3\\ 0 & otherwise\end{cases}$

$(Ans: 11/6, 11/3, 11/36, 0.553)$

(c) $f(x)=\begin{cases}\frac{3}{56}(x+2)^2 & 0\le x\le 2\\ 0 & otherwise\end{cases}$

$(Ans: 17/14, 62/35, 291/980, 0.545)$

17. A random variable X has probability density function $f(x)$ given by

$$f(x)=\begin{cases}\frac{k}{x(4-x)} & 1\le x\le 3\\ 0 & otherwise\end{cases}$$

(a) Show that $k=\frac{2}{\ln 3}$
(b) Calculate the mean and variance of X.
$(Ans: 2, 4-\frac{4}{\ln 3})$

HAZBER SAMSON

18. The continuous random variable X has p.d.f $f(x)$ given by where

$$f(x)=\begin{cases}\frac{1}{27}x^2 & 0\le x<3\\ \frac{1}{3} & 3\le x\le 5\\ 0 & otherwise\end{cases}$$

(a) Find $E(X)$
(b) Find $E(X^2)$
(c) Find Standard Deviation of X.
$(Ans: 41/12, 571/45, 1.008)$

19. A random variable X has probability density function $f(x)$ where

$$f(x)=\begin{cases}kx & 0\le x<1\\ k(2-x) & 1\le x\le 2\\ 0 & otherwise\end{cases}$$

Than find
(a) The value of *k*
(b) $E(X)$
(c) $Var(X)$
(d) $P\left(\frac{3}{4}\le X\le\frac{3}{2}\right)$
(e) The Mode
$(Ans: 1, 1, 1/6, 19/32, 1)$

20. The probability density function of a random variable X is given by

$$f(x)=\begin{cases}k(ax-x^2) & 0\le x\le 2\\ 0 & otherwise\end{cases}$$

(a) Show that $a\ge 2$ and that $k=\frac{3}{6a-8}$
(b) Given that $E(X)=1$, calculate the value of *a* and *k*.
(c) Find $Var(X)$
$(Ans: a=2, k=0.75, \text{var}(x)=0.2)$

HAZBER SAMSON